

Previous cooked up example:

Assumed that we had a
Sample of KV

$$p(i) \propto \phi(\langle k_i, \bar{a} \rangle)$$

But in a auto-regressive
generation total KV tokens
keep on increasing

Q. How do we "maintain" a
Sample in that case?

A. Reservoir Sampling.

Consider a stream of data

$$\{x_1, x_2 \dots x_t \dots\}$$

at each point t , we want to
keep a sample of size K

$$\{x_{i_1}, x_{i_2} \dots x_{i_K}\}$$

such that

$$P(x_i \text{ is in Sample}) = \frac{K}{t}$$

$$\forall t$$

"Stream": we process the
data point only once and
"throw" it away.

At time t we look at x_t

"Build a streaming algorithm
for sampling"

A.

let us say we have a sample

$\{x_i, \dots, x_{i_k}\}$ at time t

$$P(i) = \frac{k}{t}.$$

At time $(t+1)$, we got x_{t+1}

Q. Choose x_{t+1} with Prob?

$\left(\frac{k}{t+1}\right) \left[\begin{array}{l} \text{check } r \sim U[0,1] \\ r \leq \frac{1}{t+1} \end{array} \right]$

Q. If chosen, throw one of
the other sampled data
with uniform probability

$$P(x_{t+1} \text{ is in Sample at } (t+1)) = \frac{k}{t+1}$$

$$P(x_i \text{ is in sample at } (t+1))$$

$$= P(x_i \text{ is in sample at } t) \times$$

$$P(x_i \text{ is not thrown})$$

$$= \frac{k}{t} \left[\dots \right]$$

$$= \frac{k}{t} \left[\frac{1+t-k}{1+t} \cdot 1 + \dots \right]$$

$$= \frac{k}{t} \left[\frac{1+t-k}{1+t} + \frac{k}{1+t} \cdot \frac{k-1}{k} \right]$$

$$= \frac{k}{t} \left[\frac{1+t-k+k-1}{1+t} \right]$$

$$= \frac{k}{t} \cdot \frac{t}{1+t}$$

$$= \frac{k}{1+t}$$

Thus the algorithm works.

Another simple algorithm, $\therefore$ Random

key algorithm.

$\rightarrow$ Every x_i is assigned a random value $r_i \sim U[0,1]$

$\rightarrow$ At each point t , the Sample is the x_i 's with

Cont...

Highest k values.

- random key assignment gives a permutation.
- probability that any key is chosen is k/t at all times t .

→ At each time you only have to keep track of top k x_i s and their scores

At $t+1$:

$$S = \{x_i, \dots, x_{i_t}\}$$

$$R = \{r_i, \dots, r_{i_t}\}$$

$$r_{t+1} = U[0,1]$$

If $r_{t+1} < \min R$:

discard x_{t+1}

Else

Add (x_{t+1}, r_{t+1})

and discard the
min

In our previous example
we wanted to sample
proportional to some
weights.

Weighted Reservoir Sampling

Given a stream,

$$\{(x_1, w_1), (x_2, w_2) \dots\}$$

Maintain a sample of k values
s.t.

$$P(x_i \text{ in sample}) \propto w_i$$

Algo

Assign every value x_i

a random value

Computed as

$$r_i \sim U[0, 1]$$

$$key_i = r_i^{1/w_i}$$

Keep top- k keys.

lemma:

If r_1 and r_2 are $U[0,1]$,

$$X_1 = r_1^{1/\omega_1} \quad X_2 = r_2^{1/\omega_2}$$

$$\omega_i > 0$$

$$P(X_1 \leq X_2) = \frac{\omega_2}{\omega_1 + \omega_2}$$

Proof:

$$\begin{aligned} & P_r(X_1 \leq X_2) \\ &= P_r\left(r_1^{1/\omega_1} < r_2^{1/\omega_2}\right) \\ &= P_r\left(r_1 < r_2^{\omega_1/\omega_2}\right) \end{aligned}$$

$$= \int_{r_2=0}^1 \int_{r_1=0}^{r_2^{\omega_1/\omega_2}} dr_1 dr_2$$

$$= \int_0^{\omega_1/\omega_2} r_2 dr_2$$

$$= \frac{r_2^{\omega_1/\omega_2 + 1}}{\frac{\omega_1}{\omega_2} + 1} \Big|_0^1$$

$$= \frac{\omega_2}{\omega_1 + \omega_2}$$

□

How to extend
the previous
algorithm to WRS?
↳ A. Chao algorithm.

Weighted Reservoir Sampling Stream

$$\{(x_1, w_1), (x_2, w_2) \dots (x_t, w_t) \dots\}$$

Maintain a sample of size k

such that

$$P(x_i \text{ in sample at } t) \propto w_i$$

Can we apply same Algo?

x_{t+1} is selected with probability?

$$\text{Say } P_{t+1} = \frac{w_{t+1} \cdot k}{\sum w_{t+1}}$$

What is the issue?

p_{t+1} can be > 1

Q: How do we get around this? [Think]

Another Algorithm for
Simple reservoir sampling.

"random key approach."

